

I. A study was undertaken to compare the mean time spent on cell phones by male and female college students per week. Fifty male and 50 female students were selected from Midwestern University and the number of hours per week spent talking on their cell phones determined. The results in hours are shown in Table 10.6. It is desired to test $H_0 : \mu_1 = \mu_2$ versus $H_1 : \mu_1 \neq \mu_2$ based on these samples.

Table 10.6 Hours spent talking on cell phone for males and females at Midwestern University

Males					Females				
12	4	11	13	11	11	9	7	10	9
7	9	10	10	7	10	10	7	9	10
7	12	6	9	15	11	8	9	6	11
10	11	12	7	8	10	7	9	12	14
8	9	11	10	9	11	12	12	8	12
10	9	9	7	9	12	9	10	11	7
11	7	10	10	11	12	7	9	8	11
9	12	12	8	13	10	8	13	8	10
9	10	8	11	10	9	9	9	11	9
13	13	9	10	13	9	8	9	12	11

1&2.

<https://colab.research.google.com/drive/1xNjIoGq4YkzCtEV7NRZIGzAZcpGvCx0E?usp=sharing>

3. A summary of the descriptive statistics for the entire dataset.

Looking at all 100 students together, they spend an average of 9.76 hours per week on their cell phones. The lowest time reported was 4 hours, and the highest was 15 hours. Since the mean (9.76) and median (10) are close to each other, and the skewness is basically zero, the data is very balanced and looks a lot like a standard, normal distribution.

A comparison of the descriptive statistics across gender categories.

When we split the data by gender (n=50 for each group), the numbers are still similar. Male students average 9.82 hours a week, which is just a tiny bit higher than the female average of 9.70 hours. Males also had a slightly wider range of answers (from 4 to 15 hours) compared to females (from 6 to 14 hours), meaning the guys' phone habits varied just a little bit more.

Interpretation of the 95% confidence intervals for the mean (overall and by gender).

- **Overall:** We can be 95% sure that the true average cell phone time for all students at the university is somewhere between 9.37 and 10.15 hours per week.

- **Males:** We can be 95% sure that the true average for male students is between 9.21 and 10.43 hours.
- **Females:** We can be 95% sure that the true average for female students is between 9.20 and 10.20 hours.

A brief discussion of any patterns or differences observed between gender groups

Overall, male and female students use their phones pretty much the exact same amount. Even though the guys had a slightly higher average (by a measly 0.12 hours) and a bit more variety in their times, their habits are practically identical to the girls. Because the 95% confidence intervals overlap a ton, there is no real statistical difference between the two groups. In short, this means we fail to reject the null hypothesis because neither gender uses their phone significantly more than the other.

II

1.) Given:

$$X = \{2, 3, 7, 8, 10, 10, 12, 15, 17, 20\}$$

$$N = 10$$

$$\sum X = 104$$

$$\sum X^2 = 1384$$

$$\sum X^3 = 20906$$

$$\sum X^4 = 341476$$

a.) $m'_1 = \frac{\sum X}{N} = \frac{104}{10} = 10.4$

b.) $m'_2 = \frac{\sum X^2}{N} = \frac{1384}{10} = 138.4$

c.) $m'_3 = \frac{\sum X^3}{N} = \frac{20906}{10} = 2090.6$

d.) $m'_4 = \frac{\sum X^4}{N} = \frac{341476}{10} = 34147.6$

2.) Given:

$$X = \{2, 3, 7, 8, 10, 10, 12, 15, 17, 20\}$$

$$N = 10$$

$$\bar{X} = 10.4$$

$$\sum (X - \bar{X}) = 0$$

$$\sum (X - \bar{X})^2 = 302.4$$

$$\sum (X - \bar{X})^3 = 222.48$$

$$\sum (X - \bar{X})^4 = 18989.472$$

a.) $m_1 = \frac{\sum (X - \bar{X})}{N} = \frac{0}{10} = 0$

b.) $m_2 = \frac{\sum (X - \bar{X})^2}{N} = \frac{302.4}{10} = 30.24$

c.) $m_3 = \frac{\sum (X - \bar{X})^3}{N} = \frac{222.48}{10} = 22.248$

d.) $m_4 = \frac{\sum (X - \bar{X})^4}{N} = \frac{18989.472}{10} = 1898.9472$

Stirling

3) Given:

$$m_4 = 1898.9472$$

$$m'_3 = 2096.6$$

$$m'_1 = 10.4$$

$$m'_2 = 138.4$$

$$m'_4 = 34147.6$$

Formula:

$$m_4 = m'_4 - 4m'_1m'_2 + 6(m'_2)^2 m'_2 - 3(m'_1)^4$$

$$1898.9472 = 34147.6 - 4(10.4)(2096.6) + 6(10.4)^2(138.4) - 3(10.4)^4$$

$$1898.9472 = 34147.6 - 86968.96 + 89816.064 - 35095.7568$$

$$1898.9472 = 1898.9472$$

verified

III. Given:

- h be the mean

- r th moment origin: $m'_r = \frac{\sum x^r}{N}$

- r th moment mean: $m_r = \frac{\sum (x-h)^r}{N}$

$$x = (x-h) + h$$

$$m'_4 = \frac{\sum [(x-h) + h]^4}{N}$$

$$[(x-h) + h]^4 = (x-h)^4 + 4(x-h)^3 h + 6(x-h)^2 h^2 + 4(x-h)h^3 + h^4$$

Sterling

$$m'_4 = \frac{\sum (x-h)^4}{N} + \frac{\sum 4h(x-h)^3}{N} + \frac{\sum 6h^2(x-h)^2}{N} + \frac{\sum 4h^3(x-h)}{N} + \frac{\sum h^4}{N}$$

$$m'_4 = m_4 + 4h(m_3) + 6h^2(m_2) + 4h^3(m_1) + h^4(1)$$

$$m'_4 = m_4 + 4hm_3 + 6h^2m_2 + 4h^3(0) + h^4$$

$$m'_4 = m_4 + 4hm_3 + 6h^2m_2 + h^4$$

Proven

IV

1.)

$$P(-t_1 < T < t_1) = 0.90$$

$$t_{0.05, 20} = 1.725$$

$$\alpha = \frac{1-0.90}{2} = 0.05$$

$$v = 20$$

$$2.) P(T > t_1) = 0.025$$

$$\alpha = 0.025$$

$$t_{0.025, 20} = 2.086$$

$$v = 20$$

$$3.) P(T < -t_1) + P(T > t_1) = 0.01$$

$$2 \cdot P(T > t_1) = 0.01$$

$$\alpha = 0.005$$

$$v = 5$$

$$P(T > t_1) = 0.005$$

$$t_{0.005, 5} = 4.032$$

Sterling

$$4) P(T > +1) = 0.55$$

$$P(T < +1) = 1 - 0.55 = 0.45$$

$$P(T > -1) = 0.45$$

$$V = 1\%$$

$$-1 = 0.128$$